

OLIVIA NALWOGA

DATA ANALYST | DATA SCIENCE | DATA ENGINEERING

Cottage Grove, MN | nalwogaolivia93@gmail.com | Portfolio | LinkedIn | GitHub

EDUCATION

University of St. Thomas | St. Paul, MN

Expected

Master of Science in Data Science | GPA: 4.0/4.0

Relevant Coursework: Python, Cloud Computing, Database Management Systems & Design, Data Preparation & Analysis, Data Analytics & Visualization, Data Warehousing.

Makerere University | Kampala, Uganda

Bachelor's Degree in Statistics | GPA: 4.02/5.0

TECHNICAL SKILLS

Dec 2024

Programming & Data Analysis: Python, pandas, NumPy, SQL, Excel, Data Cleaning, Data Analysis, Data Visualization, Data Reporting

Statistical Methods: Statistical Analysis, Regression, LASSO, Model Validation, Cross-validation, Permutation Testing, Bootstrap Methods

Databases & Platforms: Oracle SQL Developer, Relational Database Design, Data Modeling, AWS S3, Jupyter Notebook, GitHub, Cognos Analytics, Kobo Toolbox

Enterprise Systems: Banner, SharePoint, iCIMS, Alerus, Optix

DATA SCIENCE PROJECTS

High-Dimensional Variable Selection in NHANES Health Data | Python / Statistical Learning

Apr 2026 - May 2026

- Analyzed 92,988 NHANES records and 220 variables to develop and evaluate predictive models for hemoglobin levels.
- Applied LASSO regression to reduce 14 candidate predictors to 8 key variables while preserving predictive performance and improving model interpretability.
- Validated model robustness using 1,000 permutation tests and 2,000 bootstrap resamples; compared full and reduced models using test-set MSE.

Market & Salary Analytics | Oracle SQL / Database Design

- Designed a normalized relational database with 8 related tables covering job seekers, employers, skills, job postings, applications, and salary data.
- Implemented primary and foreign keys, validation constraints, junction tables, 3 database triggers, and 2 SQL views in Oracle SQL.
- Developed analytical SQL queries to examine salary patterns by skills, education, industry, and application activity.

EXPERIENCE

Graduate Research Assistant - Computer Systems Support | University of St. Thomas

Aug 2026 - Present

- Tutor graduate Data Science students in Python, SQL, and technical course concepts; troubleshoot code and data-related problems.
- Provide technical support for PCs, lab peripherals, software, operating systems, and academic technologies in engineering labs and classrooms.
- Troubleshoot hardware, software, and networking issues and collaborate with faculty, graduate students, IT staff, and research teams.

Student Assistant, Human Resources | University of St. Thomas | St. Paul, MN

Dec 2025 - Present

- Process and verify student payroll and employment records while maintaining accuracy, confidentiality, and timely completion.
- Perform data entry, auditing, and validation using Banner, SharePoint, Optix, iCIMS, Alerus, Cognos Analytics Portal, and Excel audit reports.
- Support student employee onboarding and respond to data and process questions from staff, faculty, and students.

Assistant Planner | Uganda Human Rights Commission | Kampala, Uganda

Jun - Aug 2025

- Designed and administered surveys in Kobo Toolbox and compiled quantitative and qualitative data for organizational reporting.
- Analyzed quarterly data in Excel, conducted validation assessments, and translated findings into insights for annual reporting and planning.

Update Officer / Registration Assistant / Planning Intern | NIRA | Kampala, Uganda

Jun 2023 - May 2025

- Registered applicants and updated demographic and biometric records, emphasizing data accuracy and completeness.
- Supported community data-collection activities and maintained accurate records for registration initiatives.

CERTIFICATION

Generative AI for Students - University of St. Thomas Digital Badge, Mar 2026